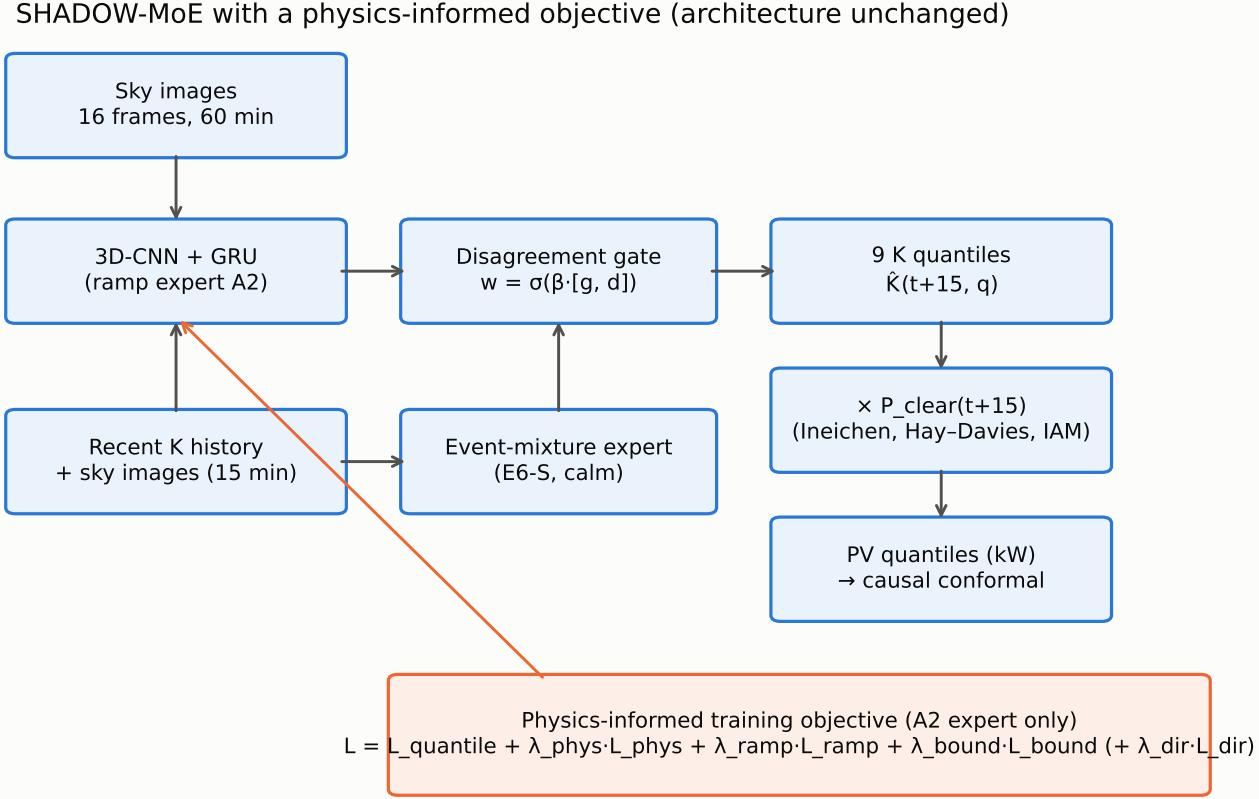
SHADOW-MoE with a physics-informed objective (architecture unchanged)
Sky images
16 frames, 60 min
3D-CNN + GRU
(ramp expert A2)
Disagreement gate
w = σ(β·[g, d])
9 K quantiles
K̂(t+15, q)
× P_clear(t+15)
(Ineichen, Hay–Davies, IAM)
PV quantiles (kW)
→ causal conformal
Recent K history
+ sky images (15 min)
Event-mixture expert
(E6-S, calm)
Physics-informed training objective (A2 expert only)
L = L_quantile + λ_phys·L_phys + λ_ramp·L_ramp + λ_bound·L_bound (+ λ_dir·L_dir)